

# A Projected Frobenius Envelope Gap for the Literal V59 Output Lane

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## Abstract

TPC-273 showed that a finite correlation margin can move across quantitative bands when a declared cutoff is changed. We next test the standard cancellation-free response: apply a Frobenius norm envelope after the same three-block Haar projection. On the locked literal V59 operator, let  $A_\perp = (I - P_3)A$ ,  $g = A\beta$ , and  $G_\perp = \|A_\perp\beta\|_2^2$ . We prove the exact finite inequality  $G_\perp \leq \|A_\perp\|_F^2 \|\beta\|_2^2 =: G_F$ . Exact rational construction and independent replay on six registered growing-cutoff scales and two kernel exponents certify  $G_F/G_\perp > 50$  and the envelope margin proxy  $m_F^2 < 1/64$  on all 12 rows. This is a quantitative, scoped insufficiency of a norm-only output proof; it is not an upper bound on the actual margin, an asymptotic counterexample, or a twin-prime theorem.

## 1 Question and claim ceiling

The residual route separates a signed scalar  $C_\perp$  from two norm lanes. If  $W_\perp = \|(I - P_3)w\|_2^2$  and  $G_\perp = \|(I - P_3)g\|_2^2$ , then the correlation margin is

$$m^2 = \frac{|C_\perp|^2}{W_\perp G_\perp}.$$

TPC-272 converted a signed saving and a margin loss into an endpoint budget; TPC-273 then demonstrated finite cutoff sensitivity. The present paper asks whether a simple output estimate can pay that missing margin cost.

Our claim ceiling is

$$\text{PROVED\_EXACT\_FINITE} \quad + \quad \text{NUMERICALLY\_CERTIFIED\_FINITE},$$

with a scoped ‘INSUFFICIENT\_SCOPED’ verdict for the tested cancellation-free route. No fixed-power credit is assigned.

## 2 The locked finite operator

For a physical scale  $N$ , put  $L = N/2$ , and let  $\beta$  be the exact source vector supplied by the released TPC-268 engine. On the prime shell  $Q < q \leq 2Q$ , its literal matrix is

$$A_{u,t} = \mathbf{1}_{u \neq t} \sum_q q K_{H,s}(u-t) \mathbf{1}_{q \nmid u} \mathbf{1}_{q \nmid t} \left( \mathbf{1}_{u \equiv t \pmod{q}} - \frac{1}{q-1} \right),$$

where  $K_{H,s}(h) = (1 + (h/H)^2)^{-s}$ . The diagonal deletion, unit masks, prime shell, beta, and four consecutive blocks are frozen; only the registered kernel exponent  $s \in \{1, 2\}$  is paired as a control.

The six growing-cutoff rows are

$$(N, H, Q) = (64, 15, 4), (96, 20, 5), (128, 24, 5), (192, 32, 6), (256, 38, 6), (384, 50, 7),$$

with comparison cutoff  $z_N = 4, 4, 4, 5, 5, 5$ , respectively. Let  $P_3$  denote the orthogonal projection onto the three declared four-block Haar contrasts and set

$$A_\perp = (I - P_3)A, \quad g_\perp = A_\perp \beta, \quad G_\perp = \|g_\perp\|_2^2.$$

All entries of  $A$ ,  $P_3$ , and  $\beta$  are rational on these rows.

### 3 The projected Frobenius theorem

**Theorem 1** (projected Frobenius envelope). *For every finite real or complex matrix  $B$  and vector  $v$ ,*

$$\|Bv\|_2^2 \leq \|B\|_F^2 \|v\|_2^2.$$

*Consequently, on every registered row,*

$$G_\perp \leq G_F := \|A_\perp\|_F^2 \|\beta\|_2^2.$$

*Proof.* For each row  $i$ , Cauchy–Schwarz gives

$$\left| \sum_j B_{ij} v_j \right|^2 \leq \left( \sum_j |B_{ij}|^2 \right) \left( \sum_j |v_j|^2 \right).$$

Summing over  $i$  yields the first inequality. Substitution of  $B = A_\perp$  and  $v = \beta$  gives the second. In particular, the theorem does not use an independence or probabilistic assumption; its cost is that it discards cancellation between the columns of  $A_\perp$ .  $\square$

For positive finite lanes define the conservative proxy

$$m_F^2 = \frac{|C_\perp|^2}{W_\perp G_F}.$$

The theorem implies  $m_F^2 \leq m^2$ . Thus a small  $m_F$  is not a counterexample to a large actual margin: it only says that this envelope cannot certify one.

### 4 Exact finite audit

The producer constructs every matrix entry and projection coefficient as a ‘**Fraction**’. It checks  $A\beta = g$  against the parent engine, computes  $\|A_\perp\|_F^2$ ,  $\|\beta\|_2^2$ , and  $G_F$  exactly, and transfers only the parent outward intervals for  $C_\perp$ ,  $W_\perp$ , and  $G_\perp$ . A separate checker repeats the matrix construction without importing the producer. A five-mutation stress audit rejects altered theorem counts, envelope values, threshold metadata, operator status, and fixed-power credit.

**Theorem 2** (registered finite gap). *On all 12 rows, the exact envelope satisfies  $G_F/G_\perp > 50$  and the transferred conservative proxy satisfies  $m_F^2 < 1/64$ . The phase intervals are separated from zero: 11 are on the negative real axis and one is on the positive real axis.*

| $N$ | $s$ | $z_N$ | lower gap $G_F/G_\perp$ | upper $m_F^2$ | phase    |
|-----|-----|-------|-------------------------|---------------|----------|
| 64  | 1   | 4     | 63.25                   | 0.001183      | negative |
| 64  | 2   | 4     | 51.35                   | 0.001197      | negative |
| 96  | 1   | 4     | 128.83                  | 0.000698      | negative |
| 128 | 1   | 4     | 237.09                  | 0.000066      | negative |
| 192 | 1   | 5     | 174.15                  | 0.000001      | negative |
| 256 | 2   | 5     | 247.09                  | 0.000001      | positive |
| 384 | 1   | 5     | 479.52                  | 0.000003      | negative |

Table 1: Selected outward finite bounds. The complete certificate has 12 rows; displayed decimals are only for readability.

*Proof.* The six scale rows are paired with  $s = 1, 2$ , giving 12 distinct keys. The certificate stores the exact rational Frobenius energy and beta energy, then divides the resulting exact  $G_F$  by the parent outward interval for  $G_\perp$ . It also divides the squared scalar interval by the positive  $W_\perp$  interval and exact  $G_F$ . The lower and upper endpoints pass the strict thresholds on every row. Independent reconstruction reproduces all exact fields and interval transfers. The phase count is a direct sign count of the stored scalar intervals.  $\square$

The gap is not a claim about a limiting sequence. It is a finite diagnostic that the rowwise norm envelope can be much looser than the actual output for the locked source. The exponent control is similarly descriptive: the six envelope ratios for the transition  $s = 1 \rightarrow 2$  range from about 0.62 to 0.63, while the actual output ratios remain distinct, so the envelope does not reproduce the source cancellation.

## 5 Route evaluation and limits

The exact inequality is a reusable structural lemma, and the finite gap is a new numerical certificate. Its negative conclusion is deliberately narrow: the cancellation-free projected Frobenius route is insufficient on the registered interface. It does not refute a signed estimate using the actual coefficient phases. In particular, the following gates remain open:

SOURCE\_LEVEL\_OUTPUT\_BOUND = OPEN\_ASYMPTOTIC  
SIGNED\_FOUR\_PACKET\_REASSEMBLY = OPEN  
ARITHMETIC\_L2 = NONE.

The finite rows also contain one positive phase, which is retained rather than removed by a sign convention. No row count is converted into a power saving, and the required strict 1/400 endpoint payment remains unpaid.

## 6 Conclusion

TPC-274 closes one low-cost proof attempt: after the physical projection, a valid Frobenius envelope loses more than a factor of 50 on every registered row and yields a margin proxy below 1/8 everywhere. The natural next question is signed output reassembly, where the coefficient phases are kept instead of discarded. That question is not answered here; the present result is a finite, reproducible obstruction to the norm-only shortcut.

## References

- [1] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford University Press, 2008.
- [2] L. Wang, “Finite Cutoff-Sensitivity Obstruction for a V59 Residual,” TPC-268 project release, 2026.
- [3] L. Wang, “A Finite Margin-Stability Matrix for the Literal V59 Residual,” TPC-273 project release, 2026.